

Katie Lester

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Research Interests

binary stars (spectroscopic, visual, eclipsing), orbital motion, stellar evolution, radial velocities, astrometry, exoplanets, planet host stars, observational astronomy: high resolution spectroscopy, long baseline interferometry, speckle interferometry, time series photometry

Education

Ph.D. in Astronomy, Georgia State University	2020
M.S. in Physics, Georgia State University	2017
B.S. in Astrophysics, Lehigh University	2014

Appointments

Visiting Assistant Professor, Mount Holyoke College	2023 – present
NASA Postdoctoral Fellow, Ames Research Center, with Dr. Steve Howell	2020 – 2023
Graduate Research Assistant, Georgia State University, with Dr. Douglas Gies	2014 – 2020
Undergraduate Research Assistant, University of Wyoming, with Dr. Chip Kobulnicky	2013

Teaching Experience

Mount Holyoke College, Instructor	2023 – present
– ASTR 100: Stars & Galaxies – introductory astronomy course for non-science majors.	
– ASTR 226: Cosmology – intermediate level course for science majors.	
– ASTR 228: Astrophysics I – introductory astronomy course for physics & astronomy majors.	
– ASTR 335: Astrophysics II – upper level stellar structure course for astronomy majors.	
– ASTR 352: Astrophysics III – upper level extragalactic course for astronomy majors.	
Georgia State University, Teaching Assistant	2014 – 2017
– ASTR 1010 – weekly labs for introductory stellar astronomy course.	
– ASTR 1020 – weekly labs for introductory extragalactic astronomy course.	

Mentorship

Undergraduate Research Students	
– Josephine Singleton (Mount Holyoke College '26)	Summer 2024 – present
– Alex Twomey (Mount Holyoke College '26)	Summer 2025 – present
– Sally Shepherd (Mount Holyoke College '27)	Summer 2025 – present
– Viktoria Sargent (Mount Holyoke College '27)	Summer 2025 – present

Professional Service

PLATO Multiple Star Working Group member	2025 – present
– assisting with target selection for GO targets and follow-up observations of binary stars	

Telescope Allocation Committee member	2022 – present
– provided science review, grading, and discussion of NOIRLab & LCO telescope proposals	
Panel Reviewer	2021 – 2024
– provided science review and grading for NASA FINESST & XRP grants	
Journal Referee	2019, 2021
– reviewed submitted manuscripts for JAAVSO and ApJS	

Department Service

Research Bootcamp Volunteer & Organizer	2024 – 2025
– co-organized and led a week-long introduction to research for undergrad summer interns	
Senior Thesis Prize Reviewer	2024 – 2025
– reviewed and graded senior astronomy theses for the Five Colleges’ Mary Irvine Prize	
Career Panel Member	2019 – 2024
– participated in several panel discussions about graduate school and career paths for undergrads	
Astronomy Peer Advising Leaders, President & Mentor	2016 – 2020
– Proposed for university funding, maintained the club’s budget, and lead monthly mentor meetings	
– Organized and lead new student orientation, professional development workshops, and mock qualifying exams for graduate students	
– Provided advice and support during monthly one-on-one meetings with junior graduate student mentees	

Outreach

Williston Observatory Open House Volunteer	2023 – present
– Operated historic reflector & lawn telescopes and discussed astronomy with the general public	
STEMPOC Mixer Volunteer	2023 – 2024
– Faculty representative at a mixer for STEM students of color, talked about influential astronomers of color and the astronomy program	
Podcast Guest	2021
– Spoke about my search for companions to exoplanet hosts on the “365 Days of Astronomy” podcast	
Hard Labor Creek Observatory Volunteer	2014 – 2020
– Operated telescopes and answered questions from the public during monthly open houses	
Solar Eclipse Event	2017
– Operated solar telescopes and engaged with the public during a solar eclipse viewing party	
Urban Life Observatory Volunteer	2014 – 2017
– Operated telescopes during on-campus observing sessions for astronomy lab students	
Girl Scout Workshop Volunteer	2014 – 2017
– Assisted with workshop activities, including building pinhole cameras and filter wheels	

First Author Publications

My full publication record is [listed on ADS](#). (9 first-author, 42 total, h-index = 16)

9. K. V. Lester, S. B. Howell, R. A. Matson, et al. [2023](#), AJ, 166, 166, “Visual Orbits and Alignments of Planet Hosting Binary Systems”

8. K. V. Lester, G. Schaefer, F. Fekel, et al. [2022](#), AJ, 164, 228, “Visual Orbits of Spectroscopic Binaries with the CHARA Array. IV. HD 61859, 89822, 109510, and 191692”
7. K. V. Lester, S. B. Howell, D. R. Ciardi, & R. A. Matson. [2022](#), AJ, 164, 56, “Determining Which Binary Component Hosts the TESS Transiting Planet”
6. K. V. Lester, R. A. Matson, S. B. Howell, et al. [2021](#), AJ, 162, 75, “Speckle Observations of TESS Exoplanet Host Stars. II. Stellar Companions at 1-1000 au and Implications for Small Planet Detection”
5. K. V. Lester, F. Fekel, M. Muterspaugh, et al. [2020](#), AJ, 160, 58, 1, “Visual Orbits of Spectroscopic Binaries with the CHARA Array. III. HD 8374 and HD 24546”
4. K. V. Lester, D. R. Gies, G. Schaefer, C. Farrington, et al. [2019](#), AJ, 158, 6, “Visual Orbits of Spectroscopic Binaries with the CHARA Array. II. the eclipsing binary HD 185912”
3. K. V. Lester, D. R. Gies, G. Schaefer, C. Farrington, et al. [2019](#), AJ, 157, 140L, “Visual Orbits of Spectroscopic Binaries with the CHARA Array. I. HD 224355”
2. K. V. Lester & D. R. Gies. [2018](#), AJ, 156, 8, “A Photometric, Spectroscopic, and Apsidal Motion Analysis of Eclipsing Binary BW Aquarii”
1. K. V. Lester, D. R. Gies, & Z. Guo. [2016](#), AJ, 152, 194, “A Young Eclipsing Binary and its Luminous Neighbors in Sh 2-252E”

Awards & Fellowships

Co-I, Massachusetts Space Grant Consortium student summer fellowships (\$13,000)	2024 – 2025
PI, NASA Award for time on the Keck Observatory (\$12,000)	2023 – 2025
PI, NASA Award for time on the WIYN Observatory (\$5,000)	2023 – 2024
PI, NASA Postdoctoral Fellowship (\$285,000)	2020 – 2023
Outstanding Advanced Graduate Student Award, Georgia State University	2020
Outstanding Second Year Graduate Student, Georgia State University	2016
Departmental Honors, Lehigh University	2014

Invited Talks

Mount Holyoke College Astronomy Club, “Journey to Becoming an Astronomer”	2024
Planetary Science Institute, “Characterizing Planet Host Binary Systems”	2023
CHARA & VLTI Science Meeting, “Binary Star Science Using Interferometry”	2022
AAS Splinter Session, “Speckle & Long Baseline Interferometry of Binary Stars”	2022
NSF virtual site visit at the CHARA Array, “Visual Orbits of Spectroscopic Binaries”	2020
CHARA Summer School, “Observing and Data Reduction with CLIMB”	2020
Agnes Scott College, “Visual & Spectroscopic Orbits of Binary Stars”	2019

Recent Contributed Talks & Posters

Binary Stars in the Space Era, “Orbits of Planet Hosting Binary Systems” (poster)	2025
AAS #245, “Visual Orbits & Alignments of Planet Hosting Binary Systems”	2025
AAS #243, “Orbits & Inclinations of Planet Hosting Binary Systems” (poster)	2024
EPRV5 Meeting, “Orbits & Inclinations of Planet Host Binaries” (poster)	2023
AAS #241, “Detection Sensitivity of Transiting Planets in Single vs Binary Stars”	2023
NASA Postdoc Symposium, “How Does Host Star Multiplicity Affect Planet Formation?”	2022
Bay Area Exoplanet Meeting, “Which Binary Component Hosts the TESS Transiting Planet?”	2022
AAS #240, “Which Binary Component Hosts the TESS Transiting Planet?” (poster)	2022
Bay Area Exoplanet Meeting, “Close Companions of TESS Exoplanet Host Stars”	2021
AAS #237, “Close Companions of TESS Exoplanet Host Stars” (poster)	2021
AAS #235, “Visual Orbits of Spectroscopic Binaries with the CHARA Array” (dissertation talk)	2020

Competitive Observing Time Awarded

Gemini Observatory – speckle imaging with ‘Alopeke & Zorro – 11 nights in total	2021B – 2025B
WIYN Observatory – spectroscopy with NEID – 9.3 nights in total	2022A – 2025B
The CHARA Array – interferometry with MIRC-X/MYSTIC – 5.5 nights in total	2024B – 2025A
Las Cumbres Observatory – spectroscopy with NRES – 9.7 nights in total	2023A, 2024B – 2025B
Keck Observatory – spectroscopy using HIRES – 0.5 nights	2023A
CTIO – spectroscopy using CHIRON – 1 night	2023A

Observing Experience

Summary: over 2600 hours of hands-on experience observing with professional ground-based telescopes

The CHARA Array – Six 1.0m telescopes – 78 nights – long baseline interferometry	2017 – 2025
Gemini Observatory – 8.1m telescope – 110 nights – speckle interferometry	2020 – 2023
WIYN Observatory – 3.5m telescope – 5 nights – multi-object spectroscopy	2022
Apache Point Observatory – 3.5m telescope – 49 nights – echelle spectroscopy	2016 – 2020
Wyoming Infrared Observatory – 2.3m telescope – 15 nights – longslit spectroscopy	2013

References

Dr. Douglas Gies
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Dr. Steve Howell
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